

**Example 6:** Find principal conjunctive normal form for  $[(pvq) \wedge \neg p \rightarrow \neg q]$

$$\begin{aligned}
 ((pvq) \wedge \neg p) \rightarrow \neg q &\equiv \neg((pvq) \wedge \neg p) \vee \neg q \text{ (Conditional-disjunction equivalence)} \\
 &\equiv \neg(pvq) \vee \neg(\neg p) \vee \neg q \text{ (1st De Morgan's Law)} \\
 &\equiv (\neg p \vee \neg q) \vee \neg(\neg p) \vee \neg q \text{ (2nd De Morgan's Law)} \\
 &\equiv (\neg p \vee \neg q) \vee p \vee \neg q \text{ (Double-negation Law)} \\
 &\equiv (p \vee (\neg p \vee \neg q)) \vee \neg q \text{ (1st Commutative Law)} \\
 &\equiv p \vee ((\neg p \vee \neg q) \vee \neg q) \text{ (1st Associative Law)} \\
 &\equiv p \vee ((\neg p \vee \neg q) \vee \neg q) \text{ (2nd Commutative Law)} \\
 &\equiv p \vee (\neg q \vee (\neg p \vee \neg q)) \text{ (1st Commutative Law)} \\
 &\equiv p \vee \neg q \text{ (1st Absorption Law)} \\
 &\equiv \neg(\neg(p \vee \neg q)) \text{ (Double-negation Law)} \\
 &\equiv \neg(\neg p \wedge \neg(\neg q)) \text{ (2nd De Morgan's Law)} \\
 &\equiv \neg(\neg p \wedge q) \text{ (Double-negation Law)}
 \end{aligned}$$

If we order the variables as  $p_1, p_2, \dots, p_n$  there are  $2^n$  minterms, which can be assigned values from 0 to  $2^n - 1$  as follows. A number  $i$  represents the minterm  $M_i$  if the binary representation of  $i$  is  $b_1 b_2 \dots b_n$  (with leading zeroes permitted) and the variable  $p_i$  is present in the nonnegated form if  $b_i = 0$  and is present in the negated form if  $b_i = 1$ .

In general, if there are minterms  $M_0, \dots, M_{2^n-1}$  a formula which is the conjunction of  $M_{i_1}, M_{i_2}, \dots, M_{i_r}$  is denoted by  $\prod_{i=1}^r M_{i_i}$ .

$$\begin{aligned}
 (p \rightarrow q) \rightarrow (q \rightarrow p) &\equiv (\neg p \vee q) \rightarrow (\neg q \vee p) \text{ (Conditional-disjunction equivalence)} \\
 &\equiv \neg(\neg p \vee q) \vee (\neg q \vee p) \text{ (Conditional-disjunction equivalence)} \\
 &\equiv (\neg(\neg p) \wedge \neg q) \vee (\neg q \vee p) \text{ (2nd De Morgan's Law)} \\
 &\equiv (p \wedge \neg q) \vee (\neg q \vee p) \text{ (Double-negation Law)} \\
 &\equiv (\neg q \wedge p) \vee (\neg q \vee p) \text{ (2nd Commutative Law)} \\
 &\equiv ((\neg q \wedge p) \vee \neg q) \vee p \text{ (1st Associative Law)} \\
 &\equiv (\neg q \vee (\neg q \wedge p)) \vee p \text{ (1st Commutative Law)} \\
 &\equiv \neg q \vee p \text{ (1st Absorption Law)} \equiv p \vee \neg q \text{ (1st Commutative Law)}
 \end{aligned}$$

$\prod$